Spring MVC Security - Custom Login Form with Bootstrap - Overview and Coding

In this lesson, we will explore how to create an elegant and engaging Bootstrap login page. If you've ever looked at your current login page and thought that it's somewhat dull or even unappealing, you are not alone. This is our "before" picture, and it's time to transform it.



What we aim to do is add some excitement, some flair to our login page. By the end of this lesson, you will have crafted a custom login page teeming with vibrant colors, sleek CSS styles, and more. Our tool of choice will be Bootstrap, which can turn our ordinary login form into something truly beautiful.

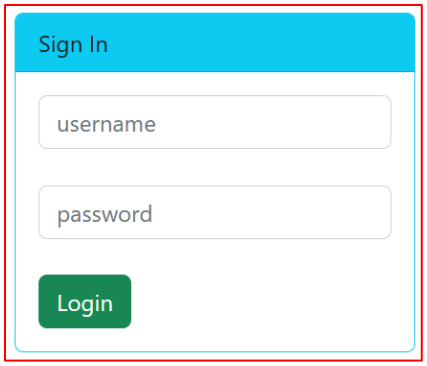
So, what exactly is Bootstrap? It is a web framework consisting of CSS styles and JavaScript, primarily designed for front-end UI development. For our project we have prepared an HTML login form using Bootstrap, essentially an HTML page with Bootstrap CSS styles applied to it. Our task will be to modify this form to integrate it with Spring Security for our project.

Here's our development process:

1. **Update the Form:** Direct the form to point to our login processing URL.
2. **Verify the Form Fields:** Check the fields for username and password.
3. **Change the Controller:** Modify our controller to utilize the new Bootstrap login form.

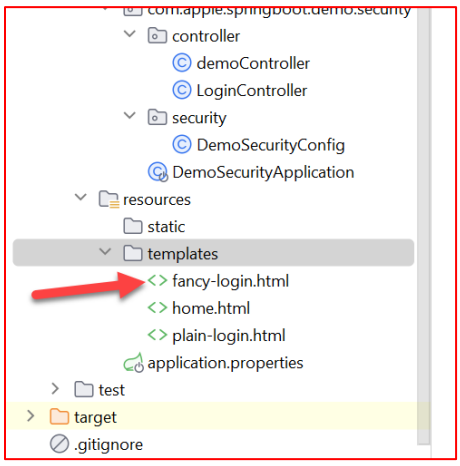
That's the plan! And remember, even if you are new to Bootstrap, there are countless free tutorials online that you can find with a simple Google search. Additionally, you may visit this website, `w3schools.com/bootstrap`, for an excellent tutorial, or refer to the Bootstrap Official Documentation available at `getbootstrap.com/docs`.

Let's get started. First, please download the starter file, which you'll find attached to this document. Feel free to pause reading here and resume once you have the file downloaded. You will find `fancy-login.html`. Let's just open this file in our web browser.



At the moment, this is merely a template, an empty canvas that we will soon bring to life. We received this basic HTML from our friend, and now we're going to copy it to our project and update it as needed.

Next, we'll copy this file and paste it into our IDE in the `source main resources templates` directory. As a checkpoint, we now have `fancy-login.html` in the right place.



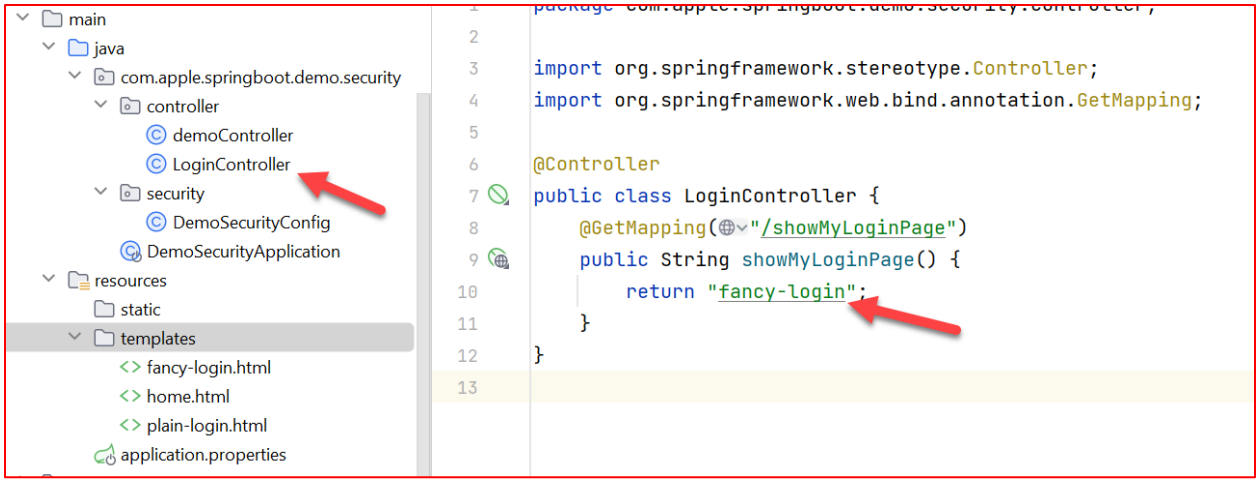
We'll now open up `plain-login.html`, and I will copy some information from it and also use it in the `fancy-login.html`. This will include the language and the XML namespace for Thymeleaf.

Our first step will be to update the form to use our login processing URL. Moving down to the form section, we'll find our login form. For the `th:action`, we will apply our login processing URL.

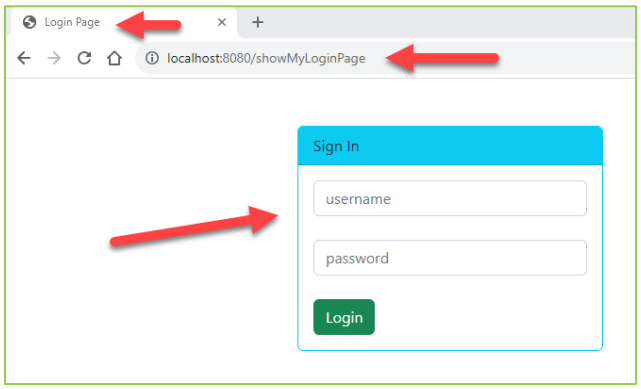
Now, as a checkpoint for your form, ensure that you authenticate the user with our login processing URL and set the method to POST. There is also a section here for checking for login errors. Again, we have written all this code before; we are merely integrating it into this new login page.

In step two, we'll verify the form fields for the username and password, ensuring they match with everything else. This will include the login submit button.

Finally, in step three, we will modify our controller to employ our new Bootstrap login form. This involves a quick copy-paste exercise to update our return statement to `fancy-login`.



With everything in place, we can run our application! Our app now sports a stylish Bootstrap login form. By attempting to log in, you'll even see a well-designed error message.



Let's go ahead and log in using the correct information for our user: username `John` and password `test123`. And there we have it; we're in our application.



Summary

In summary, we've just transformed a plain login form into a visually appealing Bootstrap login form. Well done, everyone! That's a tremendous accomplishment and an excellent example of how to create custom, engaging user experiences with Bootstrap.

```
<!DOCTYPE html>
<html lang="en" xmlns:th="http://www.thymeleaf.org">
    <head>
        <title>Login Page</title>
        <meta charset="utf-8" />
        <meta name="viewport" content="width=device-width, initial-scale=1" />
        <title>Bootstrap demo</title>
        <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.2.3/dist/css/bootstrap.min.css" rel="stylesheet"
integrity="sha384-rbsA2VBKQhggwzxH7pPCaAqO46MgnOM80zW1RWuH61DGLwZJEdK2Kadq2F9CUG65" crossorigin="anonymous" />
    </head>

    <body>
        <div class="container">
            <div id="loginbox" style="margin-top: 50px;" class="col-md-3 col-md-offset-2 col-sm-6 col-sm-
offset-2">
                <div class="card border-info">
                    <div class="card-header bg-info">
                        Sign In
                    </div>

                    <div class="card-body">
                        <div class="card-text">

                            <!-- Login Form -->
                            <form action="#" th:action="@{/authenticateTheUser}" method="POST" class="form-
horizontal">

                                <!-- Place for messages: error, alert etc ... -->
                                <div class="form-group">
                                    <div class="col-xs-15">
                                        <div>
                                            <!-- Check for login error -->
                                            <div th:if="${param.error}">

                                                <div class="alert alert-danger col-xs-offset-1 col-xs-10">
                                                    Invalid username and password.
                                                </div>

                                            </div>

                                            <!-- Check for logout -->
                                            <div th:if="${param.logout}">

                                                <div class="alert alert-success col-xs-offset-1 col-xs-10">
                                                    You have been logged out.
                                                </div>

                                            </div>

                                        </div>
                                    </div>
                                </div>

                                <!-- User name -->
                                <div style="margin-bottom: 25px;" class="input-group">
                                    <input type="text" name="username" placeholder="username" class="form-
control" />
                                </div>

                                <!-- Password -->
                                <div style="margin-bottom: 25px;" class="input-group">
                                    <input type="password" name="password" placeholder="password" class="form-
control" />
                                </div>

                                <!-- Login/Submit Button -->
                                <div style="margin-top: 10px;" class="form-group">
                                    <div class="col-sm-6 controls">
                                        <button type="submit" class="btn btn-success">Login</button>
                                    </div>
                                </div>
                            </form>
                        </div>
                    </div>
                </div>
            </div>
        </div>
    </body>
</html>
```